Shruti Vyas

Robotics & Control Engineer

RESUME_1072

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Total Experience: 14 years

PROFESSIONAL SUMMARY

Robotics & Control Engineer with 14 years of experience in robotics, using ROS 2, Python, C++, MATLAB. Experienced in autonomous navigation, control loops and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Motion Planning, C++, SLAM, Gazebo, Robot Kinematics, PLC, MATLAB, Simulink, Python, ROS 2, LiDAR, PID Control

Methods: Autonomous navigation, Control loops, Sensor fusion

PROFESSIONAL EXPERIENCE

Lead Robotics & Control Engineer | HelioNex Labs | Jan 2021 - Present

- Lowered navigation recovery events by 31% using improved obstacle handling
- Improved pick-position repeatability by 41% after calibration updates
- Reduced robotic path completion time by 18% through route and controller tuning
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Robotics & Control Engineer | Evergrid Technologies | Jan 2017 - Dec 2020

- Reduced robotic path completion time by 13% through route and controller tuning
- Validated 567 robot simulation scenarios with recorded sensor logs

Associate Robotics & Control Engineer | Foresight Industrial | Jul 2012 - Dec 2016

- Lowered navigation recovery events by 17% using improved obstacle handling
- Improved pick-position repeatability by 22% after calibration updates

PROJECTS

Autonomous Mobile Robot | Simulink, LiDAR, Gazebo, ROS 2

- Built ROS 2 nodes for localization, obstacle avoidance and route execution.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

Warehouse Mapping Trial | PLC, MATLAB, Python, LiDAR

- Benchmarked LiDAR SLAM configurations under moving obstacles.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

EDUCATION

B.Tech in Mechatronics Engineering, Horizon Institute of Engineering, 2012

Relevant Methods: Autonomous navigation, Control loops, Sensor fusion

CERTIFICATIONS

Industrial Robot Safety (2025)